STUDENT AEROSPACE CHALLENGE 2025/2026

Integrated Navigation and Orbital Awareness System

Supaero Astra Iberian Team - WP7: Reusable Propulsion / Maintenance

A. Fernández-Acero Campoamor[1], J. Soler i Pla[1], E. Valverde Sacristán[1], C. Xu[1], Á. Yuste Pubill[1]

Context

Current Systems

- ✓ Reliable navigation
- ✓ Human-rated redundancy
- × High power demand
- × Limited outage autonomy

Future Missions Need

- ✓ Energy Efficient
- ✓ Full autonomy
- ✓ GNSS resilience
- ✓ Scalable software solutions

Problem

LEO Mission's Challenges:

- *Precise Navigation Required* → Docking, Rendezvous and Collision Avoidance.
- *Continuous GNSS* → High Power Consumption.

→ **INOAS** balances Accuracy and Energy Efficiency

Objectives

- Reduce GNSS power consumption.
- Continue navigating on GNSS outages.
- Ensure safe, autonomous operation.
- Docking and rendezvous support.

INOAS Architecture & Logic

Four-layers architecture

- **PPP Engine** → precise positioning
- **UKF Estimator** → continuous PVT during GNSS-off
- **Instrument Decision** → NIS-based scheduling
- **SCAB Control** → MPC safety-critical

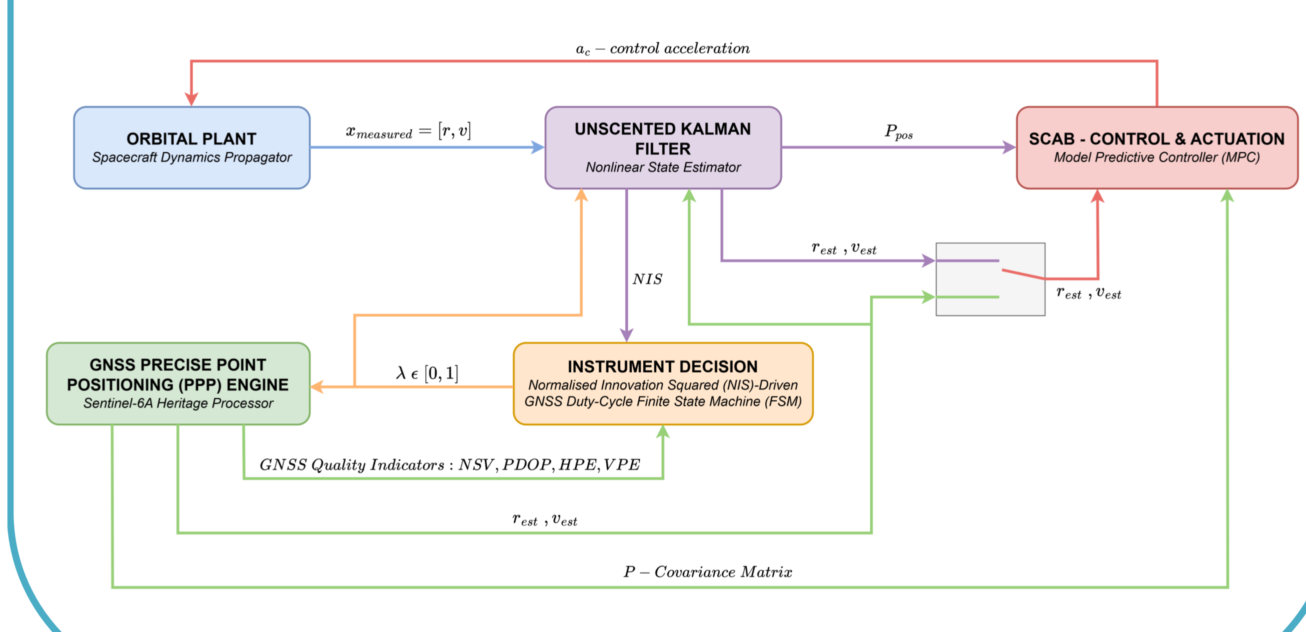


Fig 1: General layout of the Simulink simulation framework.

Core Innovation

Uncertainty-Driven GNSS Duty Cycling

When the trace of the covariance matrix (P_k) is below the threshold, GNSS is switched OFF → the UKF propagates the state autonomously.

$$d_{\text{safe},k} = d_{\text{safe},0} + \gamma\sqrt{\text{tr}(P_k^{\text{pos}})}$$

The controller becomes more conservative when GNSS is degraded.

Key Innovations

- **Software-only** → all improvements from algorithmic integration alone.
- **Heritage PPP** → validation done on real Sentinel-6A POD telemetry (Y24D011).
- **Covariance coupling** → navigation confidence directly drives control safety margins.

Validation Scenarios

Validation done using real **Sentinel-6A POD telemetry** (Dataset Y24D011) at 1347 km altitude → three representative scenarios:

Orbit	Perturbation	INOAS Response
1-2	Nominal	Initial state convergence & sensor tracking.
3	Signal Outage (< 4 satellites)	UKF autonomous estimation; GNSS-off propagation.
5	Precision lost (VPE/HPE > 5m)	Degradation proximity mode; covariance inflated margins.
7	Geometric drift (PDOP > 6)	Safe-hold / degraded control mode activated.
9	Rendezvous	MPC terminal phase ΔV = 106.6 m/s

Instrument Decision

- **λ ∈ {0,1}** → state definition: the filter propagates continuously regardless of mode.
- **GNSS_STATE** (λ=1): GNSS measurements active; switches to Kalman after $T_{GNSS,on}$ = 60 s or on Emergency.
- **KALMAN_STATE** (λ=0): UKF propagates autonomously using magnetometer + altimeter; NIS monitors filter consistency.
- **Early GNSS recovery**: if NIS ≥ 12 (filter divergence detected) or $T_{KALMAN,on}$ = 300 s and GNSS is healthy, duty-cycle timer is overridden.

$$\text{Emergency} \triangleq (N_{sat} < 4) \,||\, (\text{PDOP} > 6) \,||\, [(\text{HPE} > 5\,\text{m}) \,\&\, (\text{VPE} > 5\,\text{m})]$$

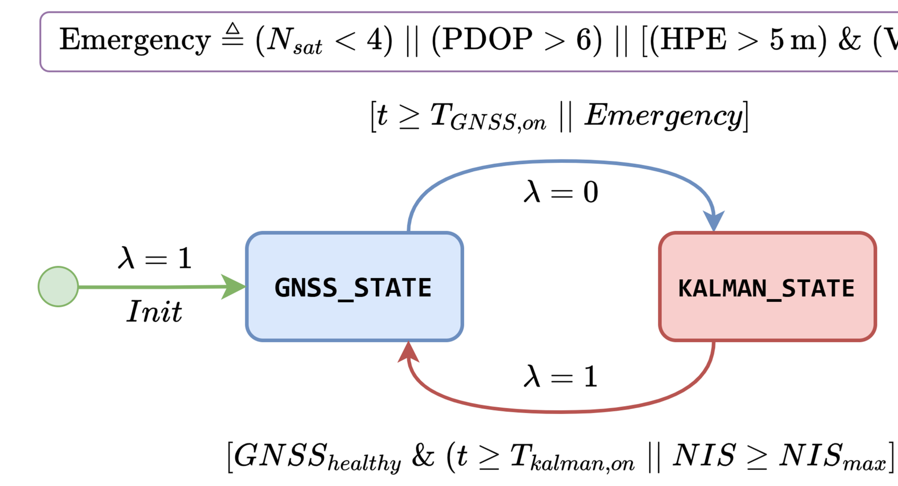


Fig 3: Instrument Decision Logic — FSM.

Spacecraft Control and Actuation Block (SCAB)

Unified green hypergolic bipropellant system: MEA fuel + HTP (≥95%) oxidizer with $Cu(NO_3)_2$ doping for sub-15 ms ignition.

Parameters	Main Thrusters (x2)	ACS Thrusters (×26)
Thrust	2,500 N each → 5,000 N total	220 N each
I_{sp}	310 s	292.1 s
Feed Cycle	Electric pump-fed	He pressure-fed
Propellant	487 kg (ΔV = 144.57 m/s)	92.2 kg (3-axis, 100 s)
Location	Aft-mounted, symmetrical	12 fwd + 14 aft blocks

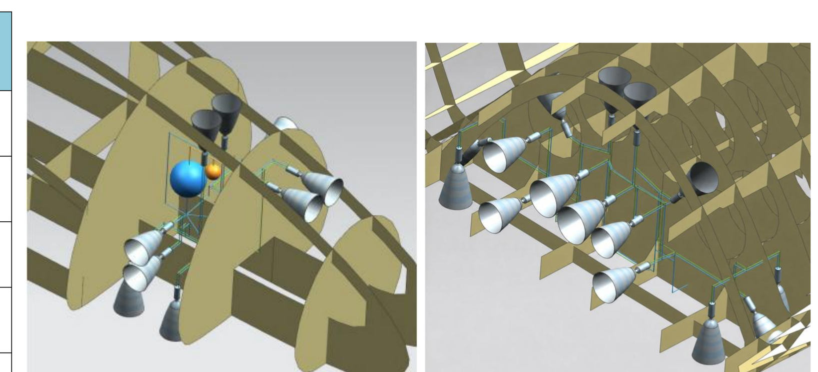
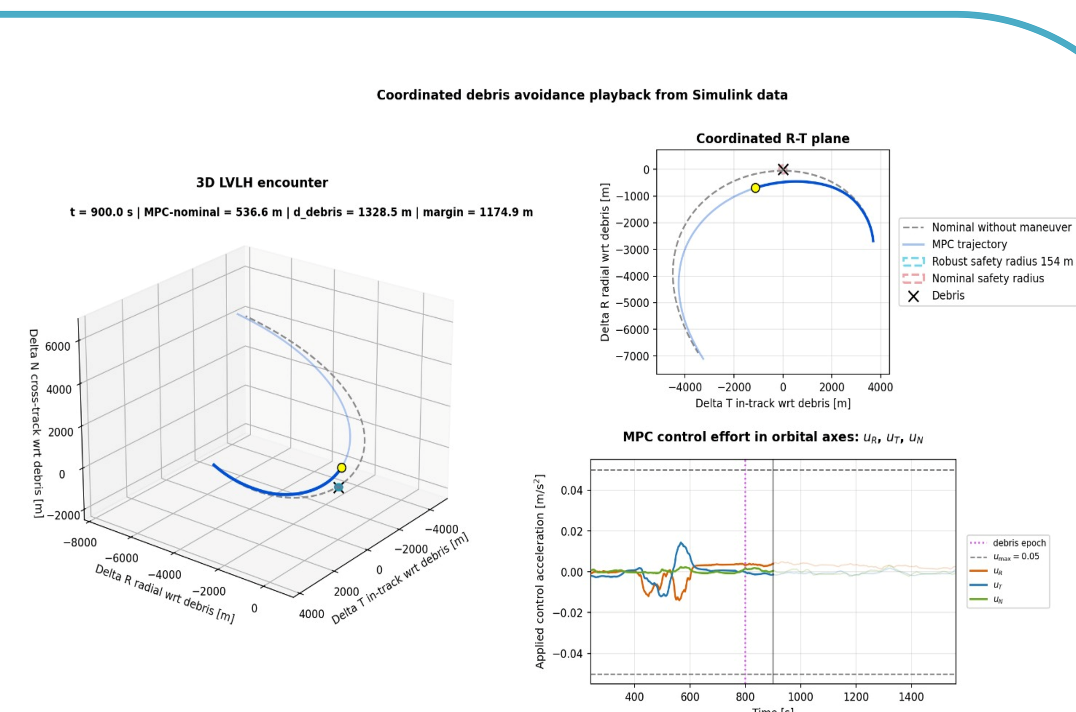


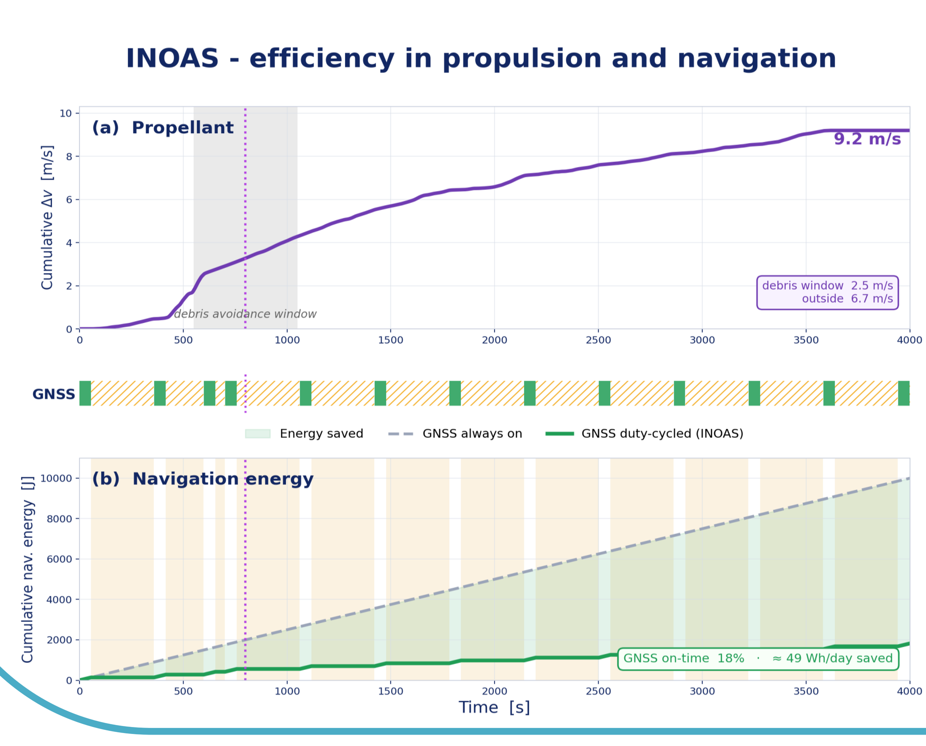
Fig 2: Forward and Aft Thrusters Placement.

Results

Debris Avoidance

- MPC maintains a minimum separation of **483.2 m** — well above the **154 m** robust safety radius.
- Control effort stays well within saturation limits: peak acceleration < 0.018 m/s² across all orbital axes (u_R, u_T, u_N)
- Without maneuver, nominal trajectory (dashed) passes through the keep-out zone — MPC intervenes autonomously before debris epoch at t ≈ 800 s





INOAS Energy efficiency

- INOAS reduces navigation energy by **~82%** vs always-on GNSS — saving ≈ **49 Wh/day** with GNSS active only **18%** of the time
- Debris avoidance maneuver costs **2.5 m/s** ΔV — total cumulative ΔV **9.2 m/s** over 4000 s
- Kalman propagates autonomously during the remaining **82%** GNSS-off time with no navigation loss

Conclusions

INOAS proves that the energy-navigation trade-off in LEO can be solved by software-only algorithmic integration.

- ✓ **82% GNSS Receiver energy saving**
- ✓ UKF position error **< 40 m** during outages - Final error **5 m** during full GNSS outage
- ✓ Covariance-driven MPC maintained debris separation at **483.2 m**

Future Work

- Complete **6-DOF model** with **attitude control**
- **Terminal docking** with **Model Predictive Control**
- **Live multi-debris tracking** via **SSA** integration
- On-board **Hardware-in-the-Loop** validation

[1]Institut Supérieur de l'Aéronautique et de l'Espace (ISAE-SUPAERO), Toulouse, France